

CV2401 Geological Engineering

Semester I: 2026 – 2027

Department of Civil & Environmental Engineering
Indian Institute of Technology Delhi

Course Instructors: Dr. Bappaditya Manna* & Dr. Deepanshu Shirole

Credits: 2 (2 - 0 - 0)

Introduction: Geology and Geological Engineering, Engineering geology of sediments, Engineering soil classification, Soil Mechanics: concepts, Geology of rock formation, Bowen's reaction series, Rock structure and classification, Attitude of rock structures and measurements, Rock mechanics: concepts, Material Models, Deposits and Landforms, Geological Mapping, Plate tectonics: Earthquakes and damage; Geophysical techniques, Geology of dam and tunnel design; Expansive soils and heaving bed- rock, slope instability classification, Groundwater in slopes, Stereonets & kinematic analysis, Slope monitoring, Tunneling concepts, Geological hazards

Course Evaluation Pattern:

Mid-Term Exam	=	50 marks
Major Test	=	50 marks

Total	=	100 marks

Time Schedule: Tuesday, Friday: 11 a.m. – 11:50 a.m.

Venue: Block VI- LT 2

Reference Books:

1. Structural Geology by M. P. Billings.
2. Structural Geology: An introduction to Geometrical Technique by Donald M Ragan, John Wiley.
3. Rutley's Elements of Mineralogy by H. H. Read.
4. Petrology by Tyrrel.
5. Hudson, J. A., & Harrison, J. P. (2000). Engineering rock mechanics: an introduction to the principles. Elsevier.
6. Luis Gonzalez de Vallejo, & Mercedes Ferrer (2011). Geological Engineering.

Important Note:

1. Students having less than 75% attendance will be given one grade less than the grade scored by them.
2. Minimum of 60% attendance is must for appearing in Major exam. Students having attendance less than 60% will be given F-grade.
3. A minimum mark for Audit Pass (NP) is 40%.